And here's one more snippet, just to demonstrate that the integer count of seconds since the *exact moment* of the epoch is truly zero:

```
var_dump($epoch->getTimestamp());
```

```
Output:
int(0)
```

Keeping Time

The need for accurate timekeeping arose from the growing industrial and financial sectors in the eighteenth and nineteenth centuries: usually an industrious person would set a clock or watch from a reference device (e.g., the clock at the Greenwich Observatory) and then spend the day visiting customers (clock and watch makers, or banks in particular). The story of the Belville family (and Ruth Belville in particular) is an interesting one; I encourage you to search for "The woman who sold time" in your preferred search engine. This method of time synchronization is not particularly conceptually different from how modern computer systems solve the problem today.

Generally speaking, all computers will track the number of seconds elapsing since an epoch. Even when powered off, a personal computer has a battery-backed chip that's counting all the time. This is great, but not all clocks are created equal: eventually they will drift out of sync with true UTC time (unless your system is equipped with something staggeringly expensive involving radioisotopes). Which brings us to…

Network Time Protocol (NTP/`ntpd`)

Just like the banks of old relied on someone to bring the correct time to them, today we have a network protocol (NTP) that allows a system to ask for the correct time from a service that is backed by something staggeringly expensive and usually involving radioisotopes. On a MacOS system you can use the `sntp` cli utility to get detailed info about how time is synchronized on your system:

```
$ cat /etc/ntp.conf
server time.euro.apple.com
```